



COMP 4300

Intro to Game Programming

Lecture 18

Introduction to Shaders

Shader Resources

- The Book of Shaders
<https://thebookofshaders.com/>
- Shader Toy
<https://www.shadertoy.com/>

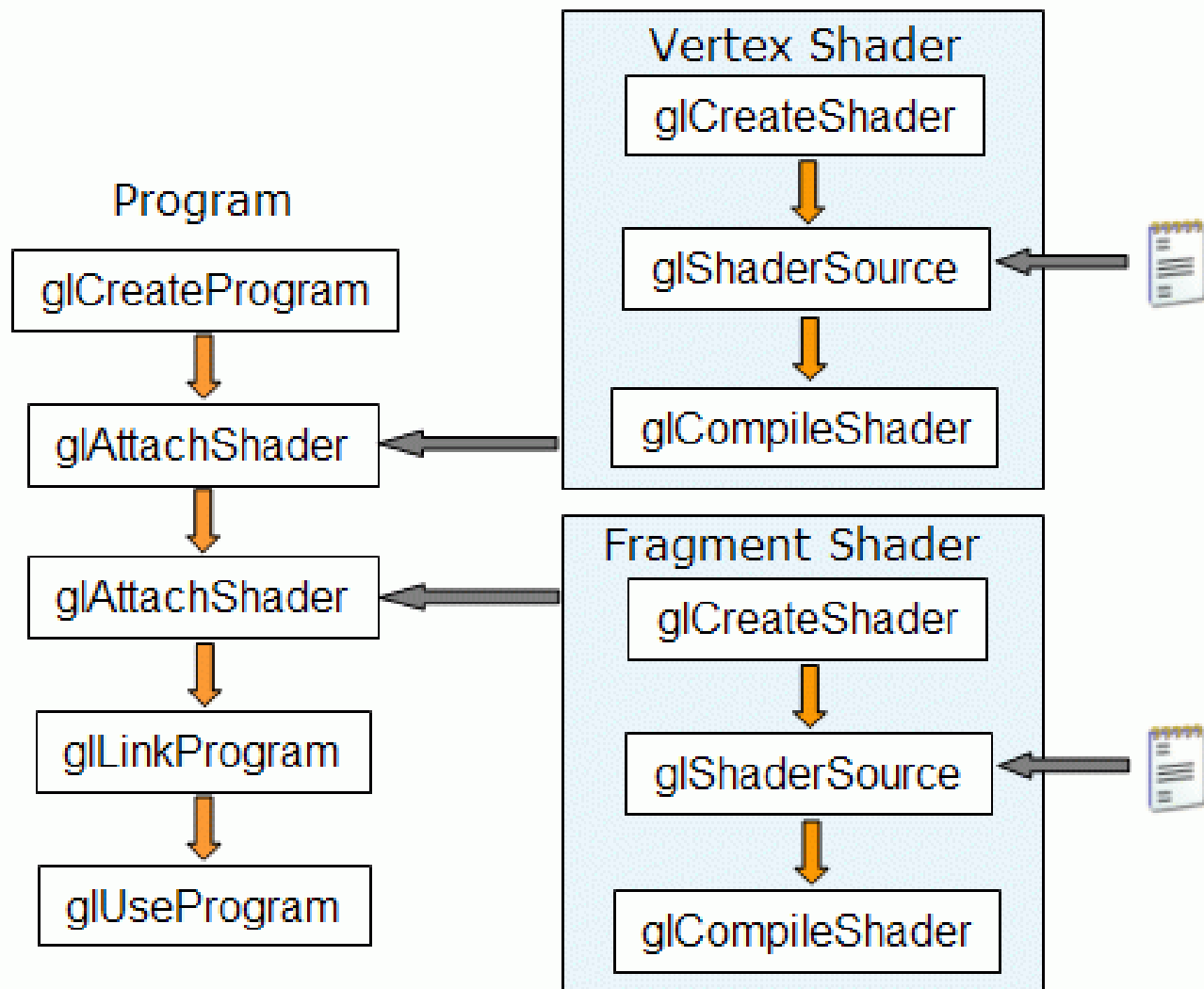
CPU vs GPU



<https://www.youtube.com/watch?v=-P28LKWTzrI>

Shaders

- Shaders are programs run on the GPU
- Shaders are programmed as a series of instructions in GLSL (GL Shader Language)
- Shader programs are compiled by the GPU
- Shaders are executed in parallel on the GPU
- Two types: Vertex and Fragment Shaders



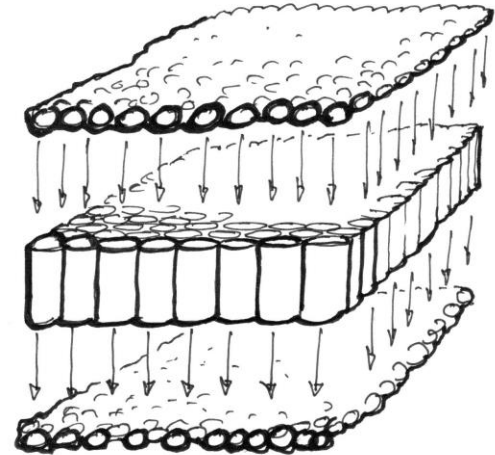
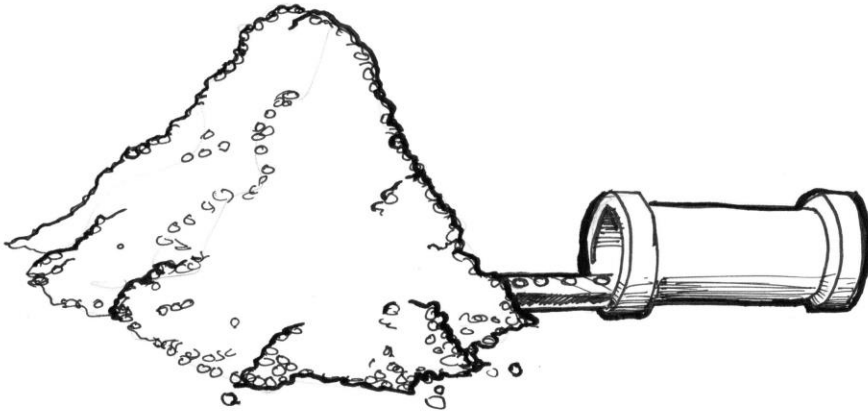
Vertex Shader

- Executed once for every vertex
- Not commonly used in 2D shaders
- Less impressive effects for 2D
- Let's skip them for now

Fragment Shaders

- Executed once per pixel (frag) being drawn
- Executed in parallel by GPU, every pixel of the screen is calculated at the same time
- Given an input pixel position and color, output a color you want to replace it
- Inputs: fragCoord
- Outputs: fragColor

CPU vs. GPU



GLSL

- Overview of GLSL

<https://www.youtube.com/watch?v=uOErsQljpHs>

`vec4 rgba = {r, g, b, a}`

`vec3 color = {r, g, b}`

`vec2 pos = {x, y}`

`vec3 gray(0.5)`

`vec4 rgba(color, a)`

Fragment Shader

<https://thebookofshaders.com/>

Fragment Shader (Textures)

```
void mainImage( out vec4 fragColor, in vec2 fragCoord )
{
    vec2 xy = fragCoord.xy / iResolution.xy;//Condensing this into one line
    vec4 texColor = texture(iChannel0,xy); //Get the pixel at xy from iChannel0
    fragColor = texColor;                //Set the screen pixel to that color

    // color part of the image
    //if (xy.x > 0.5) fragColor.r = 1.0;

    // increase red with x-value
    //fragColor.r += fragCoord.x/iResolution.x*0.5;

    // increase blue with y-value
    //fragColor.b += sin(iTime)*fragCoord.y/iResolution.y*0.5;

    // gray scale
    //float gray = (texColor.r + texColor.g + texColor.b) / 3.0;
    //fragColor = vec4(vec3(gray), 1);

    // mask out image based on grayscale value
    //if (gray < sin(iTime)) fragColor = vec4(vec3(0),1);
}
```

<https://www.shadertoy.com/new>

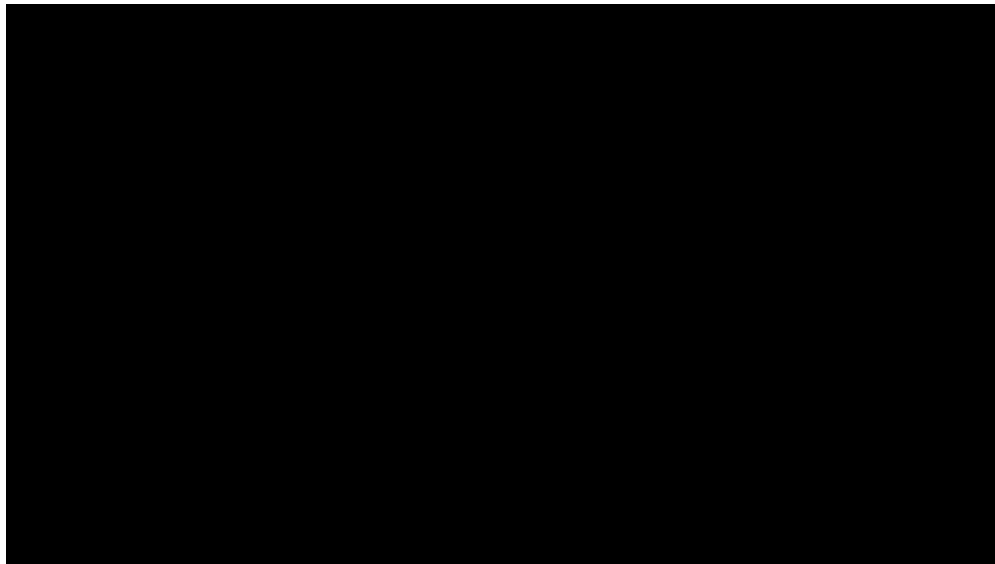
Using Shaders in SFML

```
1. sf::Shader shader;  
2. shader.loadFromFile("fragment_shader.frag",  
                      sf::Shader::Fragment)  
3. shader.setUniform("myvar", 5.0f);  
4. window.draw(whatever, &shader);
```

// Shader will be applied to whatever being drawn

<https://www.sfml-dev.org/tutorials/2.5/graphics-shader.php>

2D Shader Examples



<https://www.youtube.com/watch?v=4zqEnhxBSCc>